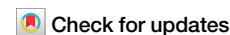




Identification of digital twins to guide interpretable AI for diagnosis and prognosis in heart failure



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Heart failure (HF) is a highly heterogeneous condition, and current methods struggle to synthesize extensive clinical data for personalized care. Using data from 343 HF patients, we developed mechanistic computational models of the cardiovascular system to create digital twins. These twins, consisting of optimized measurable and unmeasurable parameters alongside simulations of cardiovascular function, provided comprehensive representations of individual disease states. Unsupervised machine learning applied to digital twin-derived features identified interpretable phenogroups and mechanistic drivers of cardiovascular death risk. Incorporating these features into prognostic AI models improved performance, transferability, and interpretability compared to models using only clinical variables. This framework demonstrates potential to enhance prognosis and guide therapy, paving the way for more precise, individualized HF management.

Heart failure (HF), both with preserved and with reduced ejection fraction (EF), is a heterogeneous syndrome characterized by abnormal cardiac structure and function, leading to reduced cardiac output and/or elevated filling pressures at rest or with exertion^{1,2}. This highly prevalent, multifactorial condition remains associated with significant morbidity and mortality³. A major challenge in HF management is the individualization of risk and treatment, with relatively fewer adequate treatment options for those with heart failure with preserved ejection fraction (HFpEF) than for those with heart failure with reduced ejection fraction (HFrEF)^{4–6}. Due to the wide range of clinical presentations and diverse underlying mechanisms, tailoring treatments to individual needs is often difficult⁷.

To better understand and assess heterogeneity in HF, unsupervised machine learning (ML) has been applied to patient record data to identify patient subgroups with similar characteristics that have distinct prognoses^{8–12}. Recent applications integrate molecular biomarkers with clinical variables to enhance mechanistic insights into phenogroups identified through ML^{13–15}. Nonetheless, although unsupervised ML-driven analyses of large HF patient datasets have successfully identified phenogroups with different prognoses, interpreting the mechanisms behind these differences remains limited and relies on prior knowledge. Similarly, supervised learning approaches, using either traditional ML or deep learning, can be trained to robustly predict mortality in HF patients^{16–18}. However, supervised ML models used to improve performance do not easily

yield mechanistic interpretation. Recent applications use personalized computational models to create patient-specific simulations, referred to as digital twins¹⁹. These digital twins have been combined with unsupervised ML to enhance mechanistic insights into HF patient clusters²⁰ and to improve the performance of supervised ML models in predicting responses to cardiac resynchronization therapy²⁰ as well as detecting and localizing myocardial scars²¹.

We have developed a model-based, physiology-informed ML classification approach to categorize HF patients into distinct phenotypic groups, revealing that when explicit physiological prior knowledge embedded in a computational model is integrated with data-driven inference, the phenotyping process is enriched compared to a purely data-driven approach²². Here, we build on that framework utilizing digital twins—comprising optimized sets of parameters and corresponding simulations of cardiovascular system function—for HF patients in a retrospective cohort to inform both supervised and unsupervised approaches in identifying phenogroups and novel mechanistic drivers of cardiovascular death risk. We demonstrate how the integration of digital twins into AI-based analyses of patient data enhances the performance and interpretability of prognostic AI models. Finally, we show that prognostic AI models trained with digital twin features are more transferable than models trained with only clinical variables, as evaluated using an independent prospective cohort.

Results

Digital twins yield novel patient-specific insights into cardiovascular state

We conducted a retrospective review of Electronic Health Records (EHR) from 343 HF patients at the University of Michigan Health System (UMHS), gathering information on 116 clinical characteristics (Fig. 1). Among these patients, 215 were diagnosed with HFrEF, while 128 had a

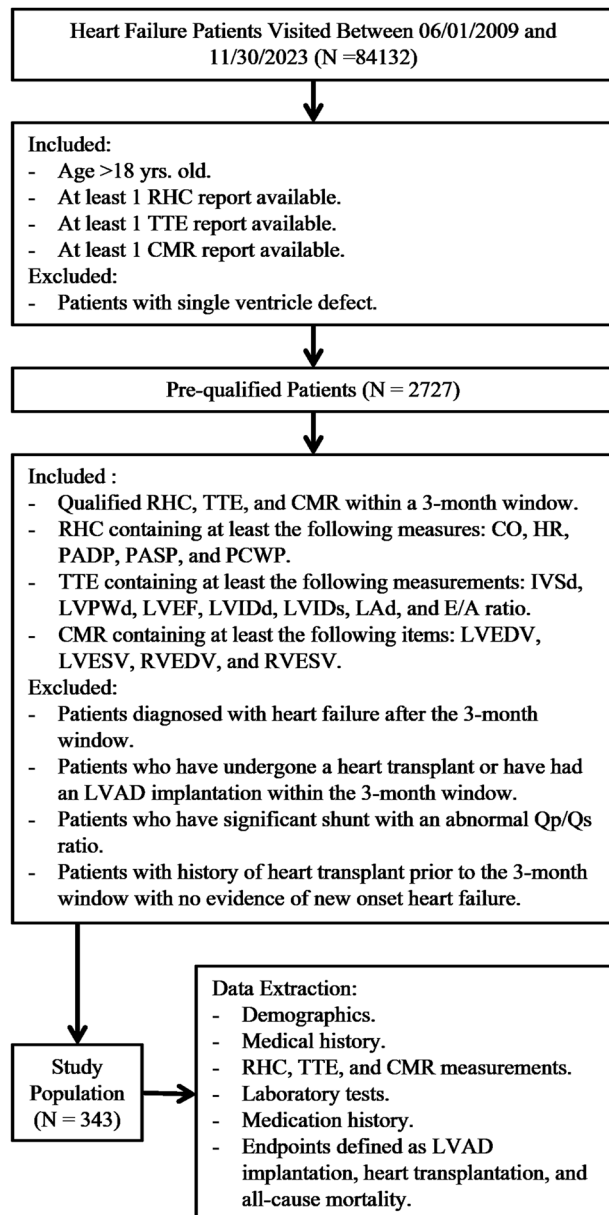


Fig. 1 | UMHS cohort discovery. The flowchart represents the UMHS cohort discovery process. The extracted data are used for digital twin identification, precision phenotyping, and building prognostic AI models. RHC Right Heart Catheterization, TTE Transthoracic Echocardiography, CMR Cardiac Magnetic Resonance Imaging, CO Cardiac Output, HR Heart Rate, PADP Pulmonary Artery Diastolic Pressure, PASP Pulmonary Artery Systolic Pressure, PCWP Pulmonary Capillary Wedge Pressure, IVSd Interventricular Septum Thickness, LVPWd Left Ventricular Posterior Wall Thickness, LVEF Left Ventricular Ejection Fraction, LVIDd Left Ventricular Internal Diameter end-Diastole, LVIDs Left Ventricular Internal Diameter end-Systole, LAd Left Atrial Diameter, LVEDV Left Ventricular End-Diastolic Volume, LVESV Left Ventricular End-Systolic Volume, RVEDV Right Ventricular End-Diastolic Volume, RVESV Right Ventricular End-Systolic Volume, Qp/Qs ratio the ratio of pulmonary blood flow (Qp) to systemic blood flow (Qs), LVAD Left Ventricular Assist Device.

baseline LVEF estimated to be $\geq 50\%$. Although these patients are labeled as HFpEF for convenience, they represent a heterogeneous group, including patients with pulmonary arterial hypertension ($N = 6$) or hypertrophic cardiomyopathy ($N = 32$), contributing to a greater degree of heterogeneity compared to the general HFpEF population. The extracted baseline characteristics are summarized in Supplementary Table 1. We further created parameterized representations of each patient using a computational model of the cardiovascular system (Fig. 2a). These representations, optimized for patient-specific resting cardiovascular function, were later found in subsequent analyses to correlate with clinical outcomes and were likely stable over time. Due to their patient-specific nature, prognostic relevance, and high likelihood of longitudinal stability, these representations serve as digital twins. In this study, each representation—comprising an optimized set of model parameters and corresponding cardiovascular simulations—constitutes the patient's digital twin. Clinical measurements used as inputs for the computational model are summarized in Supplementary Table 2. Patient-specific parameters derived from the digital twin and their meanings are provided in Supplementary Table 3.

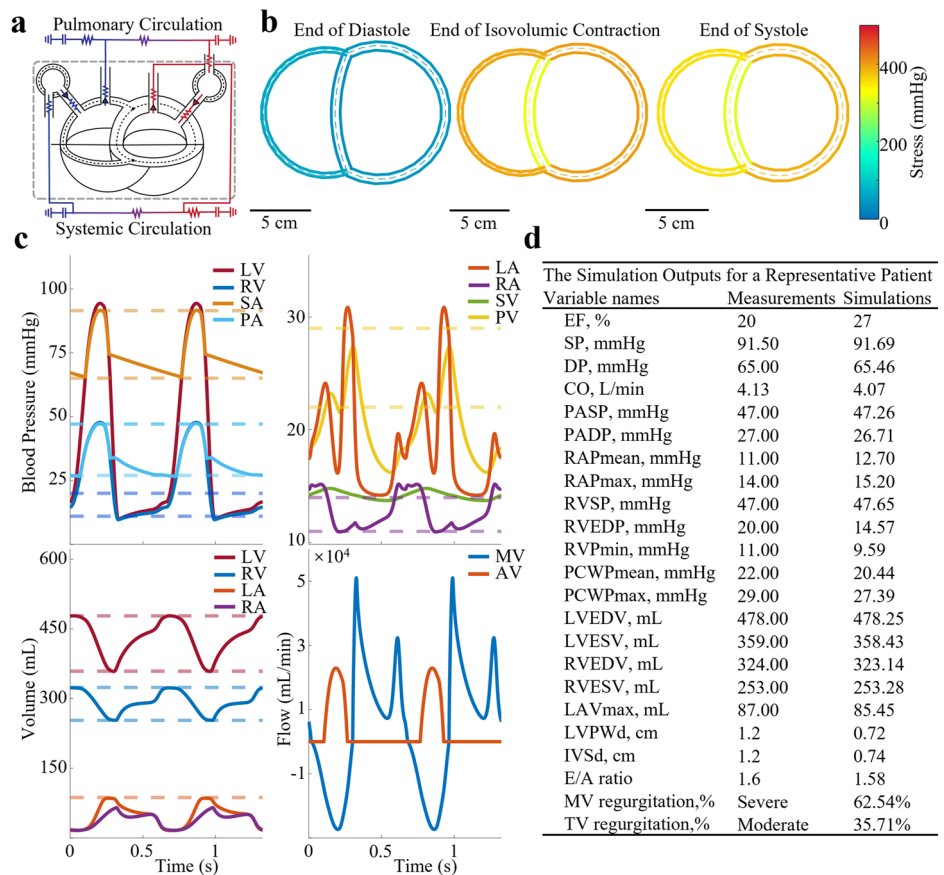
The identified digital twins represent a synthesis of anatomical measurements and volumetric flow data derived from TTE and CMR, along with pressure data from RHC. For example, simulations of a patient with HFrEF, a dilated LV, mitral insufficiency, and tricuspid insufficiency (Fig. 2b, Supplementary Movie 1) show cardiac anatomic and mechanical features consistent with the imaging and catheterization data for this individual. This individual's simulation outputs, including pulmonary and systemic pressures, mitral and aortic valve flows, and ventricular and atrial volumes (Fig. 2c), closely match the corresponding targets from clinical measurements (Fig. 2d).

The relationships between simulated and measured variables for all subjects are illustrated in Supplementary Figs. 1, 2, indicating that digital twins robustly capture the diverse cardiovascular hemodynamic characteristics present in HF patients. Moreover, in many cases, a mismatch between the data and the model reveals an inconsistency in the data rather than a shortcoming of the model. The inconsistency may arise from differences in measurement modalities. For instance, EF measured by TTE may not necessarily align with LVESV and LVEDV measured by CMR or with CO and HR obtained from RHC. Additionally, timing discrepancies between measurements and intra-patient variability can contribute to these mismatches. This is particularly relevant as we introduced a 3-month window, during which any measurement within this period was considered, while patient phenotypes may have changed over time. In addition, the digital twin provides a model-based estimate of key hemodynamic measurements sometimes included in the RHC report. Simulation outputs show a correlation with RHC report estimates (Supplementary Fig. 2). Similar to EF measurements, model-based outputs are likely more accurate than estimates in the RHC report, which do not correct for data inconsistencies and are based on a less comprehensive picture of the patient compared to the digital twin representation.

Digital twins have been further applied to interpret cardiovascular physiology in HF patients. All compliances were initially standardized by dividing them by body surface area (BSA), and all resistances were standardized by multiplying them by BSA to eliminate the influence of patient size. Subsequently, all functional parameters were normalized against corresponding parameter values derived from 20-year-old healthy subjects, with female patients compared to an ideal healthy female and male patients to an ideal healthy male (Supplementary Fig. 3). This process yielded relative values: values less than or greater than 1 represent a decreased or increased parameter value relative to the canonical patient, respectively.

Consistent with current understanding, both LV and RV contractility are lower in HFrEF than in HFpEF (Fig. 3a). Notably, HFpEF subjects in this cohort also tend to exhibit reduced LV contractility compared to normal subjects, while a significant portion of HFpEF patients show increased RV contractility (Fig. 3a). Although HFpEF patients are primarily characterized clinically by diastolic dysfunction, numerous studies have confirmed that a significant proportion also exhibit impaired systolic function^{22–24}.

Fig. 2 | Digital twins identification. **a** The modified TriSeg heart model is integrated with a lumped parameter model to simulate closed-loop hemodynamics. **b** Model-predicted short-axis heart geometry is illustrated for a representative HFrEF patient at three phases during one cardiac cycle. The color represents myofiber stress in the wall. **c** Key simulation outputs are compared to data for the same patient. Dashed lines represent measurements from RHC, TTE, and CMR, while solid lines represent the simulation outputs. **d** The table compares model outputs to clinical measurements for the example patient. Valve regurgitation degree is reported as a categorical variable in the TTE reports and as regurgitation fraction in the simulations. TriSeg three-wall segment, MV mitral valve, AV aortic valve. Other abbreviations are as previously defined at Fig. 1.



Additionally, HFpEF is associated with elevated left heart filling pressures, which are closely linked to pulmonary hypertension²⁵ and subsequent compensatory increases in right ventricular contractility^{26,27}. The digital twin approach applied here provides a rigorous quantification of individual RV and LV contractility as well as effective passive material property markers.

Examining ventricular power output through simulated pressure-volume loops (Supplementary Fig. 4) reveals that both LV and RV power outputs are lower in HFrEF than in HFpEF. Specifically, LV power output in HFrEF patients is lower than in normal subjects, while in HFpEF patients, LV power output remains preserved. Additionally, RV power output is significantly increased in both HFrEF and HFpEF (Fig. 3a). The passive stiffness of both LV and RV, as quantified through the digital twins approach, is elevated in HFrEF and HFpEF compared to normal subjects, with LV passive stiffness being significantly higher in HFrEF than in HFpEF (Fig. 3a). The surface areas of the LV, RV, and septum in HFrEF are significantly larger than those in HFpEF and normal subjects, indicating a greater degree of dilation in HFrEF hearts (Fig. 3b). In contrast, the free wall volume does not differ substantially, suggesting that while the heart enlarges, wall thickness decreases, consistent with the pathology of HFrEF (Fig. 3b). Interestingly, pericardial constraint is markedly elevated in both HFrEF and HFpEF compared to normal subjects, with a greater increase observed in HFpEF, potentially implicating pericardial involvement in HFpEF etiology²⁸ (Fig. 3c).

In systemic circulation, the compliance, resistance, and transmural resistance of systemic arteries in both HFrEF and HFpEF are comparable to those in normal subjects (Fig. 3d). However, HFrEF exhibits slight but statistically significant increases in both compliance and resistance compared to HFpEF (Fig. 3d). The compliance of systemic veins in both HFpEF and HFrEF is lower than that in normal subjects, with HFrEF showing significantly higher compliance than HFpEF (Fig. 3d).

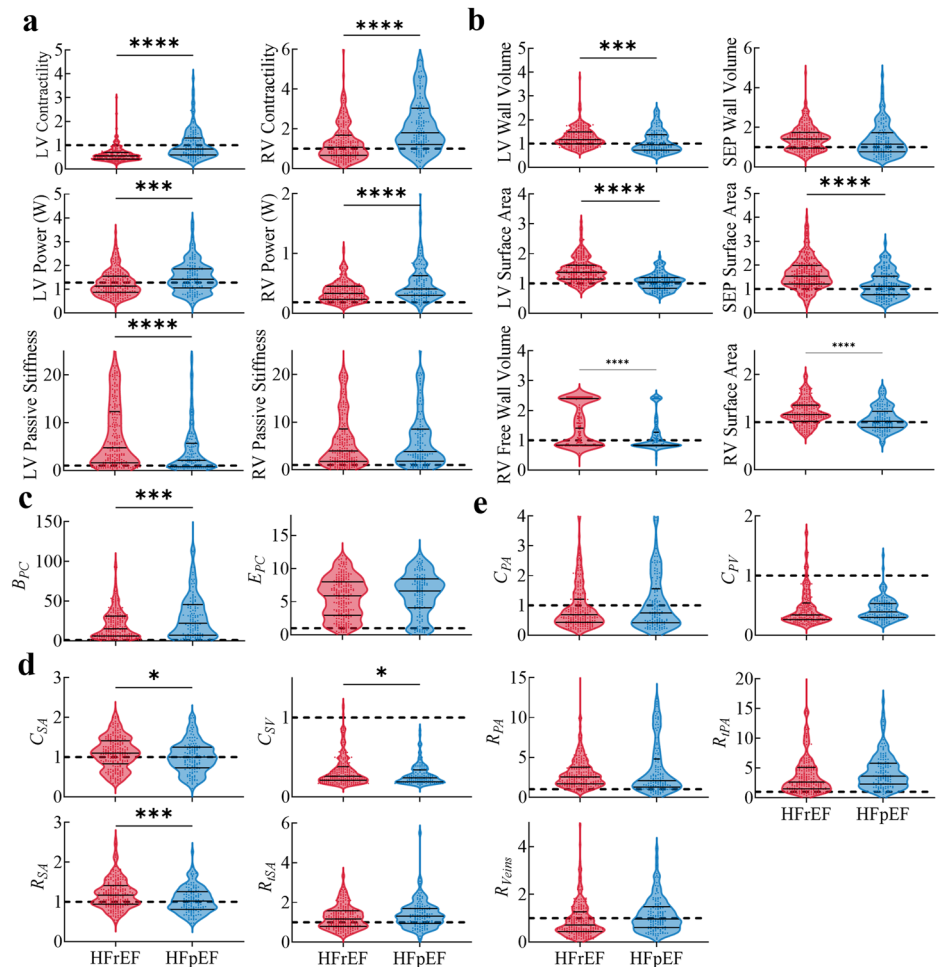
In this digital twin analysis, identical resistance values were assumed for the pulmonary and systemic veins, resulting in a single value for venous resistance. As a result, venous resistance in HFrEF and HFpEF shows no significant differences and is comparable to that in normal subjects (Fig. 3e). In pulmonary circulation, the compliance of both pulmonary arteries and veins is lower than in normal subjects, while the resistance and transmural resistance of the pulmonary arteries are elevated in both HFpEF and HFrEF (Fig. 3e). No significant differences in pulmonary circulation functional parameters are observed between HFpEF and HFrEF (Fig. 3e).

To evaluate the robustness of parameter optimization, we conducted 100 optimization runs for the representative patient shown in Fig. 2. The probability density distributions of the parameter estimates are shown in Supplementary Fig. 5. Additionally, three optimization runs were performed for 69 patients, using both minimal targets and all available targets. Supplementary Figs. 6, 7 highlight the variability in parameter estimates across independent runs. While some parameters exhibit minimal uncertainty, the three parameters with the highest run-to-run variability using all available targets were the compliance of pulmonary arteries, systemic arteries, and septum wall volume, with average correlations of estimates from independent runs of $r = 0.7784, 0.8169, 0.8135$, respectively. Using minimal targets, the parameters with the greatest variability were the resistance of veins, mitral valve, and septum wall volume, with average correlations of $r = 0.8169, 0.8172, 0.8458$, respectively.

Digital twins reveal novel phenogroups representing distinct etiologies

Figure 4a and b show the 2D principal component projection of the space of functional parameters from digital twins and the corresponding PCA loadings. The PCA loadings reveal the parameter differences that underlie three phenogroups: Phenogroup 1 is characterized by patients with high vessel compliance and/or low vascular congestion; Phenogroup 2 is

Fig. 3 | Distributions of model parameter values for HFrEF and HFpEF groups. The data are presented as violin plots with medians and interquartile ranges. Parameter values are reported relative to control representations of baseline healthy male and female. Each dot represents an individual patient from the UMHS cohort, with red indicating HFrEF and blue indicating HFpEF. Functional parameters are organized as: heart mechanical properties (a), heart geometry (b), pericardium constraint (c), systemic circulation (d), and pulmonary circulation (e). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.



characterized by patients with high pulmonary resistance and elevated pericardial constraint; and Phenogroup 3 is characterized by patients with dilated and hypertrophic hearts and elevated myocardial passive stiffness. A hybrid approach combining k-means and hierarchical clustering was applied to identify the final clustering results. Phenogroup 1 includes 61 HFrEF and 22 HFpEF patients; Phenogroup 2 includes 63 HFrEF and 88 HFpEF patients; and Phenogroup 3 includes 91 HFrEF and 18 HFpEF patients (Fig. 4c). There are no significant demographic differences across the phenogroups.

A modified 3D volcano plot was used to visualize features that show significant differences between phenogroups (Fig. 4d, Supplementary Movie 2). Each dot in the plot represents a feature, where gray dots indicate insignificant differences, and colored dots indicate significant differences. Red indicates that variables progressively increase from phenogroups 1 to 2 to 3, while blue indicates that variables progressively decrease from Phenogroup 1 to 2 to 3. Features where Phenogroup 2 exhibits the highest value are colored orange, with the color gradient shifted towards red and green based on proximity to Phenogroups 1 and 3. Features where Phenogroup 2 exhibits the lowest value are colored purple, with the color gradient shifted towards pink and navy blue based on proximity to Phenogroups 1 and 3 (Supplementary Fig. 8). Representative features from each color group are labeled and graphically compared between phenogroup populations (Supplementary Fig. 9).

All significant features, including clinical characteristics and digital twins, are visualized using heatmaps (Fig. 4e–g). Heatmap variables are listed based on graded changes across phenogroups, as captured by the 3D volcano plot. Variables are manually ordered, starting with those lowest in Phenogroup 1, followed by those in Phenogroup 2, and then Phenogroup 3.

Consistent with the PCA loadings of the functional parameters (Fig. 3a), Phenogroup 3 has the most extreme heart geometric parameters, associated with significant hypertrophy and dilation; Phenogroup 2 shows the highest pericardial constraint and mitral valve stenosis, and although pulmonary vascular resistance does not appear directly, it is reflected by the highest burden of pulmonary hypertension (PH). Phenogroup 1 exhibits heart geometry similar to that of Phenogroup 2 but has the highest vessel compliance and the lowest pericardial constraint (Fig. 4e, f).

Hemodynamic and heart morphology measurements, along with corresponding simulation outputs, were assessed across phenogroups. These variables were categorized into two groups: functional and morphological abnormalities. Functional abnormalities were defined by elevated LV filling pressures (indicated by PCWP, LVEDP, E/A ratio, and E/e') and increased RV filling pressures (indicated by elevated RAP and RVEDP), as well as PH (evidenced by higher PASP, PADP, and RVSP). Morphological abnormalities are characterized by hypertrophy and heart enlargement, indicated by EF, LVEDV, LVESV, RVESV, LVIDd, LVIDs, IVSd, LVPWd, and LV and RV mass. Phenogroup 3 exhibits both functional and morphological abnormalities, while Phenogroup 2 shows functional abnormalities with generally normal heart morphology. In contrast, Phenogroup 1 has lower LV and RV filling pressures, lower pulmonary pressures, and relatively normal heart morphology (Fig. 4e, g).

Since the clustering results are dependent on digital twin identification, we performed clustering on the results from three independent optimization runs for the 69 patients using both minimal targets and all available targets. The results indicate an average coefficient of variation of 10.87% for parameter values across optimization runs. Additionally,

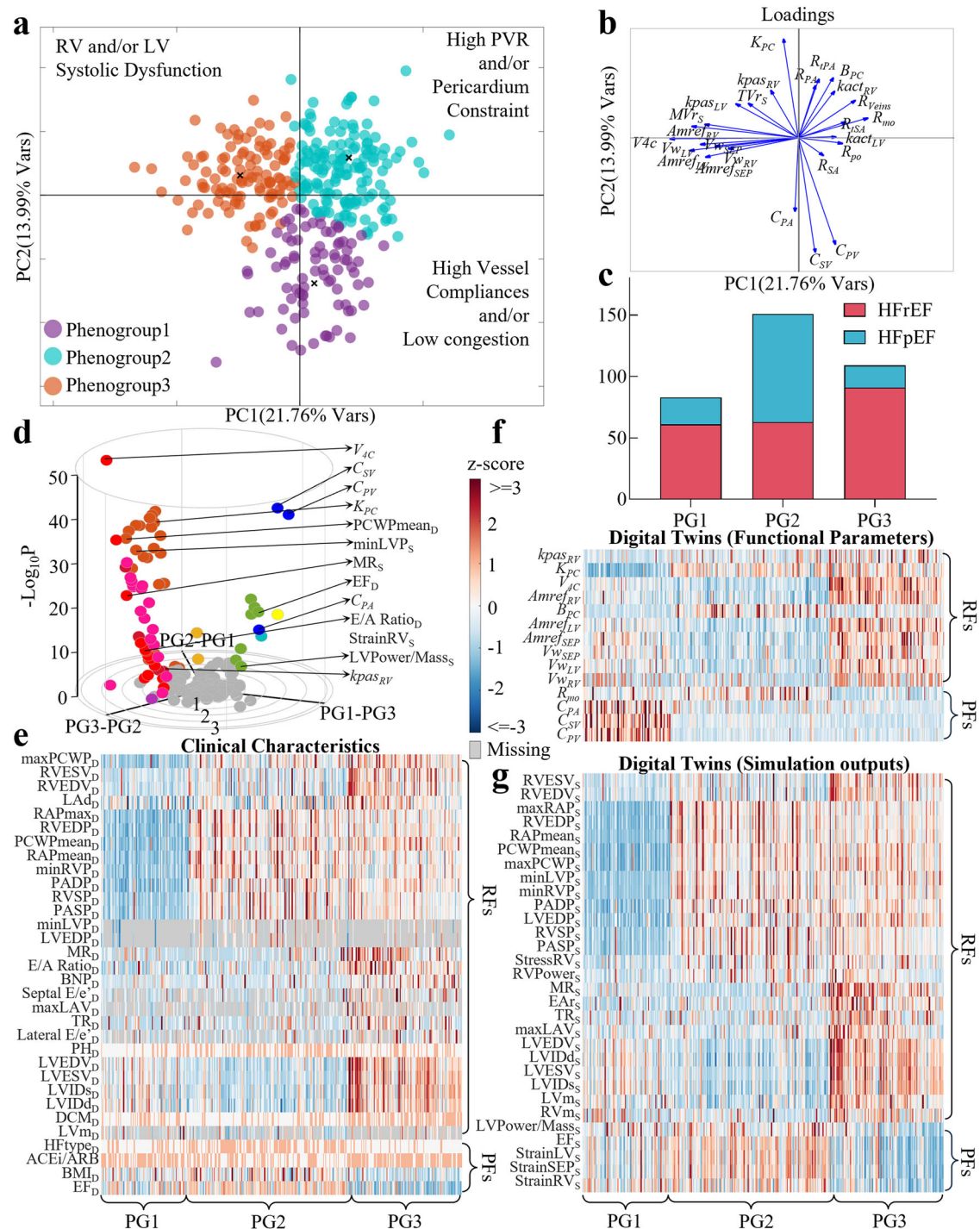


Fig. 4 | Precision phenotyping of heart failure patients. **a** Clustering results based on digital twins from UMHS cohort. Each dot represents a patient. **b** PCA loadings for PC1 and PC2. **c** Bar plot showing the distribution of diagnoses across phenogroups. **d** 3D volcano plot of clinical characteristics and digital twin features. Each dot represents a baseline feature. **e–g** Heatmaps showing significant differences in

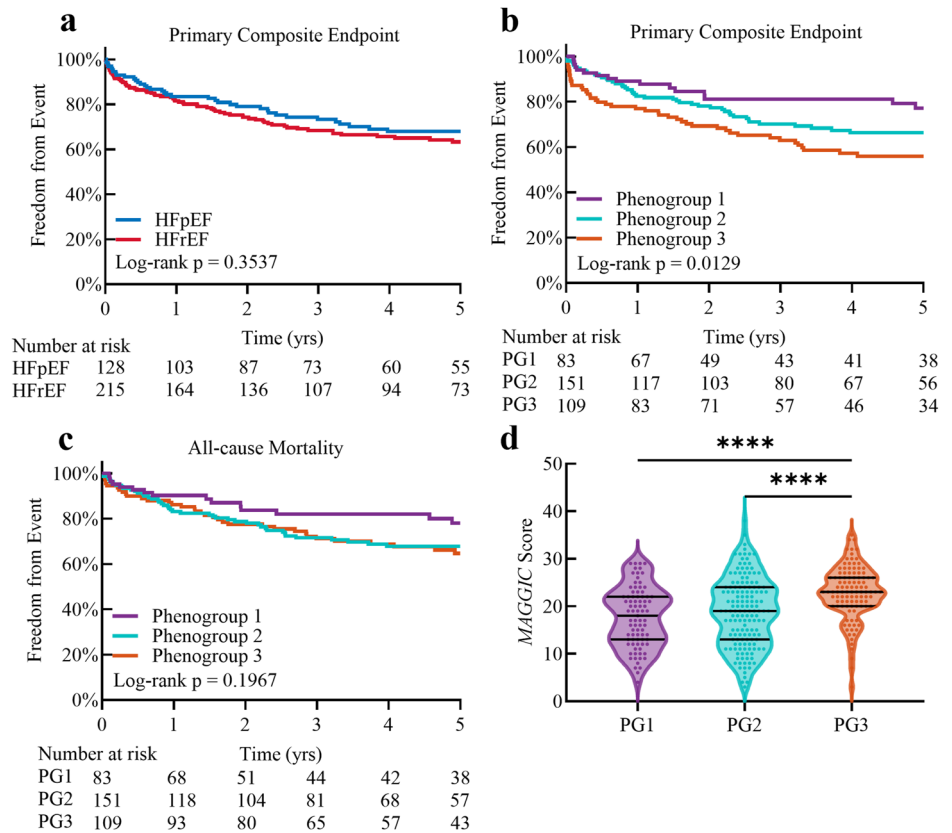
clinical characteristics, functional parameters, and simulation outputs across phenogroups. Each row represents a baseline feature and each column represents a patient. PCA principal component analysis, PC principal component, PVR pulmonary vascular resistance, PG phenogroup, RF risk factor, PF protective factor. $_D$ features from clinical data, $_S$ features from simulation outputs.

as shown in Supplementary Fig. 10, while the same patient is not always projected to the exact same location in the parameter space across runs, the projected positions consistently remain in close proximity to the original locations. Depending on the individual run and data targets used, there are between 3 and 7 out of the 69 patients who are classified into a different cluster compared to the original results. All inconsistently clustered patients are positioned near the borderline between clusters.

Phenogroups identified by digital twins have distinct prognoses

Among the 343 patients in the UMHS cohort, over a 5-year follow-up period, 107 (31.2%) reached the primary composite endpoints. These included 94 all-cause deaths (27.4%), 16 LVAD implantations (4.7%), and 8 heart transplantations (2.3%). The freedom from the primary composite endpoint shows no significant difference between HFrEF and HFpEF (Fig. 5a). However, there is a distinct pattern of association between the phenogroups identified by digital twins and the prognostic outcomes

Fig. 5 | Differences in prognostic outcomes across phenogroups. **a** Kaplan-Meier plot for the primary composite endpoint in the UMHS cohort, stratified by diagnosis. Kaplan-Meier plot for the **(b)** primary composite endpoint and **(c)** all-cause mortality in the UMHS cohort, stratified by phenogroups. **d** Differences in *MAGGIC* scores across phenogroups. The primary composite endpoint includes all-cause mortality, LVAD implantation, and heart transplantation.



(Fig. 5b, Supplementary Fig. 11). Specifically, the survival probability within 5 years for the primary composite endpoint is highest in Phenogroup 1, followed by Phenogroups 2 and 3 (80.7%, 69.5%, and 58.7%, respectively, $p = 0.0129$) (Fig. 5b). With regard to all-cause mortality, Phenogroup 1 also shows the highest survival probability (81.9%), whereas Phenogroups 2 and 3 have similar survival probabilities (70.9% and 67.9%), resulting in a non-significant difference across the three phenogroups ($p = 0.1967$) (Fig. 5c). The *MAGGIC* score, widely used for predicting all-cause mortality in HF patients²⁹, was also calculated for these three phenogroups. Phenogroup 3 has the highest *MAGGIC* score, while Phenogroups 1 and 2 have comparable scores, which are partially consistent with the Kaplan-Meier survival differences (Fig. 5d). Outcome differences were assessed not only for primary composite endpoints and all-cause mortality but also for LVAD implantation and heart transplantation. Similar associations between phenogroups and outcomes were observed with LVAD implantation ($p < 0.001$) (Supplementary Fig. 11a). However, no significant association was found between phenogroups and heart transplantation, likely due to the limited number of patients reaching this outcome (Supplementary Fig. 11b). Prognostic differences across phenogroups, observed in the overall HF cohort, were also apparent in the HFpEF subgroup ($p = 0.0314$) but were not significant in the HFrEF subgroup (Supplementary Fig. 11c). This reduced significance may be due to the majority of HFpEF patients clustering into Phenogroup 2, which limits statistical power.

The multivariable Cox proportional hazards model was employed to assess the association between phenogroups and clinical outcomes using Phenogroup 1 as the reference group. Well-established HF prognostic factors were selected as covariates²⁹, while those involved in the digital twin identification process were excluded to avoid confounding effects. Phenogroup 3 is significantly associated with a higher risk of the primary composite endpoint [hazard ratio (HR) 2.72, 95% confidence interval (CI) 1.53–5.06] and all-cause mortality (HR 1.98, 95% CI 1.08–3.80) (Table 1). Phenogroup 2 initially appears to be associated with a higher risk of the primary composite endpoints and all-cause mortality, as it shows higher

rates for both primary composite endpoints (30.5% vs. 19.3%) and all-cause mortality (29.1% vs. 18.1%) compared to Phenogroup 1. However, after adjustments for covariates, these associations are not statistically significant (Table 1). Phenogroup 3 was significantly associated with a higher risk of LVAD implantation (HR 11.87, 95% CI 2.27–218.8) compared to Phenogroup 1, but not with heart transplantation (Supplementary Table 4).

To interpret the differences in phenogroup outcomes, we identified risk and protective factors among features, based on whether or not they are increased or decreased in value or frequency in Phenogroup 1 compared to 3 (Fig. 4e, f, g). Most of the risk and protective factors from clinical characteristics are well-established in relation to HF prognosis. Risk factors include elevated left- and right-sided filling pressures, increased B-type natriuretic peptide (BNP) levels, abnormal heart geometry, as well as PH and atrioventricular valve insufficiencies. Protective factors include higher body mass index (BMI), better EF, and more frequent use of ACEi and ARBs (Fig. 4e). Beyond these clinical characteristics, the functional parameters of digital twins identify additional risk and protective factors, some of which are difficult or impossible to measure clinically. Risk factors include increased RV passive stiffness and pericardial constraint, while protective factors include preserved pulmonary and systemic vascular compliance and mitral valve resistance (Fig. 4f). Simulations from digital twins show substantial overlap with clinical risk factors and also identify novel risk factors like RV power output and RV free wall maximum stress, along with protective factors such as the LV power/mass ratio³⁰ and maximum strain of the LV, RV, and septum (Fig. 4g).

Digital twins enhance predictive AI performance

We further extended our analysis by using digital twin features in a prognostic ML model for predicting primary composite endpoints. The random survival forest (RSF) predictive model, selected for its performance^{16,18}, was trained exclusively using data from the UMHS cohort. To ensure consistency and facilitate external validation, the features included in the training process were selected based on their availability in both the UMHS

Table 1 | Association between phenogroup membership and the risk of different adverse clinical outcomes using a multivariable Cox proportional hazard analysis

	Phenogroup 1 (n = 83)		Phenogroup 2 (n = 151)			Phenogroup 3 (n = 109)		
	Events	Reference	Events	HR (95% CI)	P	Events	HR (95% CI)	P
Primary composite endpoint ^a	16 (19.3%)	Ref.	46 (30.5%)	1.11 (0.63–2.06)	0.73	45 (41.3%)	2.72 (1.53–5.06)	0.001
All-cause mortality	15 (18.1%)	Ref.	44 (29.1%)	1.14 (0.63–2.16)	0.68	35 (32.1%)	1.98 (1.08–3.8)	0.03

Outcomes were adjusted for age, sex, smoking status, race, history of diabetes, hypertension, chronic obstructive pulmonary disease (COPD), baseline creatinine levels, usage of angiotensin-converting enzyme inhibitors (ACEi), angiotensin II receptor blockers (ARBs), and beta blockers, as well as for whether the patient had a first diagnosis of heart failure within the past 18 months.

CI confidence interval, HR hazard ratio.

^aA composite endpoint includes all-cause mortality, LVAD implantation, or heart transplantation.

Table 2 | Selected variables for random survival forest

Clinical Characteristics (n = 15)	Digital Twins (n = 20)	Combined (n = 21)	Extended MAGGIC score (n = 5)
Age	<i>Compliance of Systemic Veins</i>	Age	<i>MAGGIC score</i>
Hemoglobin	<i>Compliance of Pulmonary Veins</i>	Hemoglobin	E/A Ratio
Creatinine	<i>Constant Pericardium Constraint</i>	Creatinine	LV Inner Diameter (end diastole)
BNP	RVEDP	Cardiac Output	Mean Mitral Valve Pressure Gradient
Max RA Pressure	<i>Resistance of Veins</i>	LV Posterior Wall Thickness	<i>Constant Pericardium Constraint</i>
Cardiac Output	Tricuspid Valve Regurgitation Fraction	Interventricular Septum Thickness	
LVESV	<i>Resistance of Pulmonary Valve</i>	E/A ratio	
LV Posterior Wall Thickness	<i>Resistance of Mitral Valve</i>	<i>Compliance of Systemic Veins</i>	
E/A Ratio	LV Inner Diameter (end diastole)	<i>Resistance of Mitral Valve</i>	
LVEDV	LVEDP	Mean Mitral Valve Pressure Gradient	
LV Inner Diameter (end diastole)	<i>Resistance of Tricuspid Valve (backwards flow)</i>	Peak Pulmonary Valve Pressure Gradient	
<u>Diabetes Mellitus</u>	Mean Mitral Valve Pressure Gradient	<i>Pulmonary Vascular Resistance</i>	
RVSP	Peak Pulmonary Valve Pressure Gradient	<u>End Stage Renal Disease</u>	
<u>End Stage Renal Disease</u>	E/A ratio	<u>Diabetes Mellitus</u>	
PASP	<i>Pulmonary Vascular Resistance</i>	<i>Resistance of Pulmonary Valve</i>	
—	PADP	Maximum Stress of Septum	
	DBP	NYHA Class	
	<i>Exponential Pericardium Constraint</i>	RVEDV	
	LVESV	Maximum Strain of LV	
	<i>Transmural Resistance of Systemic Arteries</i>	LV inner diameter (end diastole) Minimum RV pressure	

Italics represents functional parameters from digital twins, bold represents simulation outputs from digital twins, and underlining represents comorbidities. Normal text without any formatting represents other clinical characteristics.

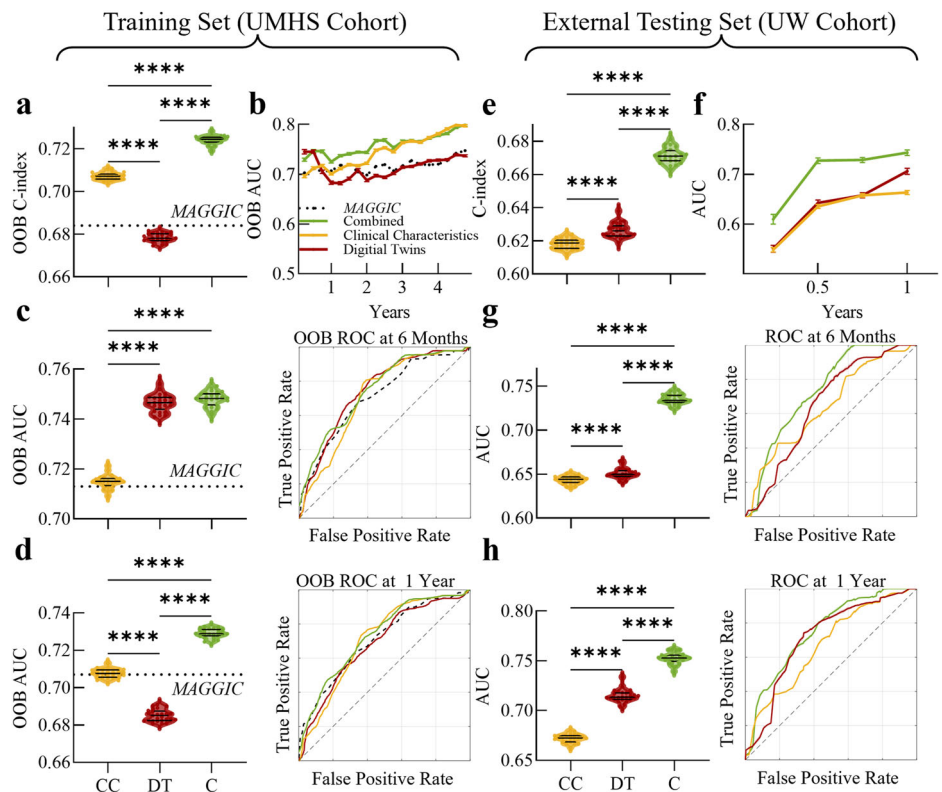
and UW cohorts (an independent cohort from the University of Wisconsin-Madison). The model was externally validated using data from the UW cohort, where only the shared features were utilized. The selected features for RSF input, ranked by variable importance, are shown in Table 2. Since the stability of features from simulations also affects the confidence in the prognostic model, we evaluated the effects of parameter uncertainty on simulation outputs used as input features for the prognostic model (Supplementary Fig. 12). The results indicate that the simulation outcomes are also very stable.

The out-of-bag (OOB) C-index of the RSF model based on combined features is the highest (0.724), compared to models using only clinical characteristics (0.707) or digital twins (0.678). For reference, the C-index for the MAGGIC score is 0.684 (Fig. 6a). Time-dependent ROC analysis shows that the RSF model based on combined features has the highest predictive power over the entire time range, with an OOB integrated area under the curve (iAUC) of 0.744, compared to 0.721 for clinical characteristics and 0.719 for digital twins (Fig. 6b). Although the prognostic ML model used

here is designed for risk prediction, we found that the predictive power varies between the short term and long term when assessed using time-dependent AUC. Therefore, we converted the ML model into a binary classification problem at specific time points (6 months, 1 year, 3 years, and 5 years). The RSF model based on digital twins demonstrates the highest predictive power within the first year, with OOB AUCs for clinical characteristics, digital twins, and combined features of 0.715, 0.747, and 0.748, respectively, at six months (Fig. 6c), and 0.708, 0.685, and 0.729, respectively, at one year (Fig. 6d), indicating that the information from digital twins enhances short-term predictions.

These findings are externally validated using the UW cohort. Since, in the UW cohort of 86 patients, only two experienced death and none underwent LVAD implantation or heart transplantation, we used the trained RSF models to predict the composite endpoint of all-cause mortality and rehospitalization. The external validation confirms that the RSF model based on combined features has the highest C-index (0.671), outperforming models using either clinical characteristics (0.618) or digital twins (0.626).

Fig. 6 | Digital twins improve prognostic AI performance. The prognostic AI, referred to here as the RSF model, is trained using clinical characteristics, digital twins, and their combination from the UMHS cohort. The RSF model was evaluated using (a) OOB C-index and (b) time-dependent AUC. The time-dependent ROC curves and corresponding AUCs for (c) 6 months and (d) 1 year are shown. The AI's performance was also externally validated using the UW cohort, with evaluations based on (e) the C-index and (f) time-dependent AUC. Time-dependent ROC curves and corresponding AUCs for (g) 6 months and (h) 1 year are also displayed. Data are presented as violin plots or mean $\pm$ SD. RSF random survival forest, OOB out-of-bag, AUC area under the curve, ROC receiver operating characteristic, CC clinical characteristics, DT digital twins, C combination of CC and DT.



Compared to the training set, RSF models based on combined features or digital twins maintain relatively stable predictive power, while the model based solely on clinical characteristics shows a greater decrease in predictive power for the validation cohort compared to the training data (Fig. 5e). The time-dependent ROC results are consistent with these conclusions, with iAUCs of 0.613, 0.623, and 0.690, respectively (Fig. 6e). In terms of 6-month and 1-year predictions, the RSF model based on combined features shows the highest performance, with AUCs of 0.735 at six months and 0.753 at one year. The predictive power of the digital twins-based model surpasses that of the clinical features-based model (Fig. 6g, h). We also validated the model using only HFpEF patients. While the combined model continued to show the highest performance, the overall performance decreases for the HFpEF-only validation group ($n = 32$, Supplementary Fig. 13). Additionally, in the UW cohort, two HFpEF patients who reached the all-cause mortality endpoint consistently had higher risk scores than the surviving HFpEF patients, regardless of whether predictions were based on clinical features, digital twins, or the combination (Supplementary Fig. 14).

We also trained and evaluated RSF models using the *MAGGIC* score combined with additional features from digital twins (Extended *MAGGIC* score) based on the UMHS cohort only. By adding four features from digital twins (Table 2), the OOB C-index increased from 0.684 to 0.731 (Fig. 7a). The model's predictive power remained stable throughout the follow-up period, with the iAUC improving from 0.710 to 0.768 (Fig. 7b). At 6 months, the AUC for the RSF model incorporating additional features reached 0.788, compared to 0.713 for the *MAGGIC* score alone (Fig. 7c), and at 1 year, these values were 0.756 and 0.707, respectively (Fig. 7d). Among all the tested RSF models, the Extended *MAGGIC* score achieved the best OOB performance, although it lacked external validation due to the UW cohort's insufficient information to calculate the *MAGGIC* score. Nonetheless, this may still suggest a mix of linear and non-linear relationships between the predictors and outcomes. Since some of the results above were obtained by converting the survival prediction problem into a binary classification problem at specific time points, we summarized the confusion matrix metrics for endpoints at 6 months, 1 year, 3 years, and 5 years in Supplementary Tables 5–8.

In the Extended *MAGGIC* score, the features E/A ratio LV inner diameter (end diastole) are available both as raw measures and as model predictions. (The mean mitral valve pressure gradient is missing in 80% of patient records and thus largely represents a model prediction not available in the raw data.) The importance of digital twin features in the Extended *MAGGIC* score may be evaluated by repeating this analysis using only raw clinical data. Using features selected exclusively from raw data (Supplementary Table 9) results in worse performance compared to incorporating digital twin features in the Extended *MAGGIC* score (Supplementary Fig. 15).

A conventional Cox proportional hazards model was also developed. Feature selection was conducted using Lasso-Cox regression with 10-fold cross-validation, as detailed in Supplementary Table 10. Using the UMHS cohort as the training set, the C-indices for Cox models based on clinical characteristics, digital twins, and combined features were 0.777, 0.669, and 0.789, respectively. The corresponding iAUC values were 0.807, 0.707, and 0.829 (Supplementary Fig. 16a). However, in external validation, the performance of all Cox models declined, with C-index values dropping to 0.624, 0.680, and 0.670, respectively, and iAUC values decreasing to 0.606, 0.674, and 0.662 (Supplementary Fig. 16b).

Discussion

HF is a highly heterogeneous and complex condition, and even at the individual level, patient care generates vast amounts of clinical data. This sheer volume of data makes it challenging to capture the key features of a patient's condition, complicating efforts for individualized management. By implementing a mechanistic computational model, we created digital twins that provide an integrated assessment of each patient's cardiovascular physiology. These digital twins offer mechanistic insights at the individual level and, through unsupervised machine learning, reveal three distinct phenogroups linked to different clinical outcomes. This approach yields interpretable insights into the underlying mechanisms of these phenogroups: Phenogroup 3 includes patients with severe RV and/or LV systolic dysfunction; Phenogroup 2 consists of patients with high pulmonary

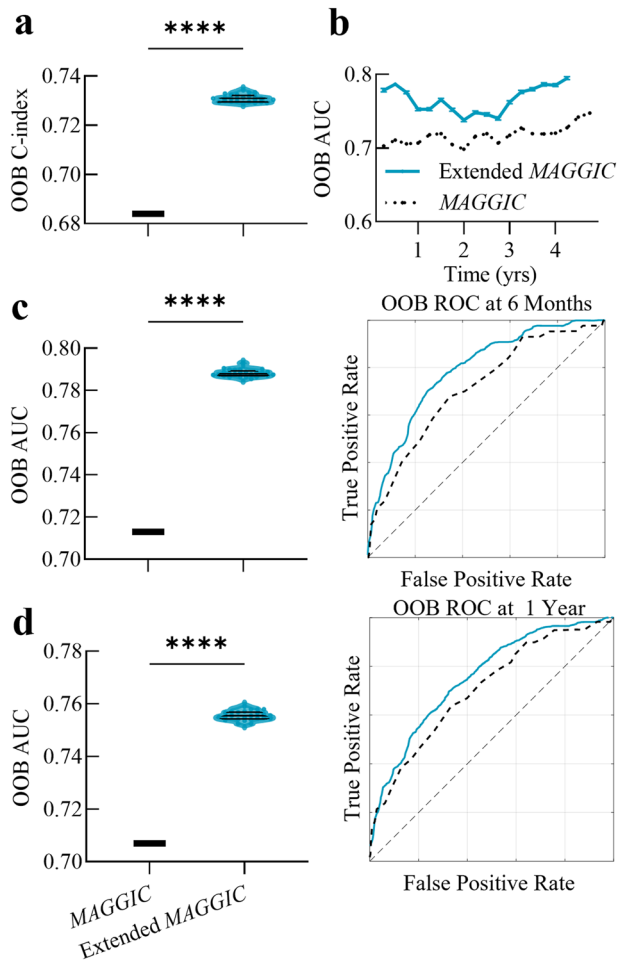


Fig. 7 | Extended MAGGIC score. The Extended MAGGIC score combines the traditional MAGGIC score with additional features derived from digital twins. Differences in (a) the OOB C-index and (b) time-dependent AUC between the Extended MAGGIC score and the original MAGGIC score are illustrated. Time-dependent ROC curves at (c) 6 months and (d) 1 year, along with the corresponding AUCs, are also shown. Data are presented as violin plots or mean $\pm$ standard deviation from the UMHS cohort. Abbreviations are as defined above.

vascular resistance or significant pericardial constraint; and Phenogroup 1 comprises patients with high vascular compliance and/or low congestion. Functional differences across these phenogroups reveal baseline risk and protective factors. Furthermore, results from supervised ML suggest that digital twins can complement and improve the performance of prognostic AI models (Fig. 8).

This simulation-based approach offers several additional advantages. First, it can address inconsistencies and inaccuracies in clinical measurements, which are relatively common. Clinical data often exhibit intra- and inter-patient variability, modality differences between measurements, and discrepancies in the timing of data collection. By incorporating all available data, the model achieves an optimal fit that balances these inconsistencies and can be used to impute missing data (Fig. 4e, g). Second, it estimates functional parameters that are otherwise impossible to measure directly, providing a more complete and accurate representation of the patient's disease state than raw clinical data alone. In addition, the digital twin framework has the potential to empower simulations of patient-specific prognoses and therapy outcomes. Ultimately, this approach has the potential to enhance the ability to focus on the most critical aspects of a patient's condition, leading to individualized care and management.

Phenomapping in HF, using ML techniques to identify distinct subgroups among HF patients^{8–15,22}, faces several key limitations. The data

sources are often chosen based on ease of access and prior knowledge, leading to significant variability across studies. Phenogroups identified in different studies often differ substantially in both number and composition. While certain subgroups may share some clinical characteristics across studies, the classification boundaries are often unclear, with considerable overlap between groups in different studies³¹. Additionally, the heterogeneity of HF as a syndrome is usually overlooked, and studies tend to under-explore important mimickers, such as amyloidosis, which can obscure the phenomapping results³². Moreover, many studies lack hemodynamic measurements, which are crucial, especially in the early stages of HFpEF. These challenges highlight the need for more comprehensive data integration, inclusion of under-explored variables like hemodynamics, and a focus on discovering novel insights beyond existing clinical knowledge.

Our digital twin hybrid with unsupervised ML helps to overcome some of these limitations. Here, phenomapping is driven by simulations matched to hemodynamic and heart geometric measurements. Moreover, our approach avoids using dimensionality reduction methods, which, while powerful, often result in less interpretable outcomes. Instead, our computational model maps patient data onto physiological functional parameters. Because physiological knowledge is embedded in the simulation framework, it enables the identification of novel phenogroups and risk factors, providing more interpretable insights than those available from purely data-driven approaches.

Most previous phenomapping studies lack longitudinal data, limiting their ability to capture trends in key variables and potentially resulting in missing important nuances in phenotype classification. While the current study may also face this limitation, we believe that, unlike many clinical variables that fluctuate rapidly, our digital twin signals are relatively stable over time. Therefore, despite not using longitudinal data, we expect the phenogroups derived from our digital twins to remain consistent and stable. In the UMHS cohort, however, we identified seven patients with two phenotyping time points and observed their phenogroup transitions. Although only three of the seven patients remained in their original phenogroups, most transitioned from higher-risk to lower-risk groups (e.g., from Phenogroup 3 to 2 to 1), with the exception of one patient who transitioned from 1 to 2. Notably, all seven patients survived beyond five years, aligning with our interpretation of the phenogroup's association with the risk of the primary composite endpoint (Supplementary Fig. 17).

The robustness and reliability of phenomapping results depend on how robust the digital twins are. We have demonstrated that the optimization process yields relatively stable results, yet with substantial uncertainty in some parameters. Moreover, uncertainty in the input data itself affects parameter identification. For a given patient's digital twin, the model may be sensitive to the different types or amounts of data available. We tested a subgroup of 69 patients from the UMHS cohort who have left heart catheter (LHC) measurements, computed their corresponding digital twins, and then compared them to the same 69 patients but without LHC measurements as input. The results show that the availability of LHC data has a limited impact on parameter optimization and subsequent phenomapping (Supplementary Fig. 10). However, a more general concern is whether measuring the same or similar variables using different methods affects the digital twin results. Nevertheless, even though uncertainty and bias in input data influence digital twin results, as long as a consistent strategy is applied, uncertainty and bias in input data should not cause additional variation when conducting statistical analyses or machine learning.

A limitation of the current phenomapping study is the lack of an external validation set. Although we cannot independently cluster UW patients since the number of UW patients is substantially smaller than that of UMHS patients, we assigned UW patients to the existing clusters and projected the UW patients onto the UMHS parameter PCA space (Supplementary Fig. 18). Specifically, 41 UW patients belong to Phenogroup 1, 37 patients to Phenogroup 2, and only 8 patients to Phenogroup 3. This distribution is consistent with our current cluster interpretation. Phenogroup 3 includes most HFpEF patients, whereas the UW cohort does not have HFpEF patients. A significant proportion of the UW cohort has PH,

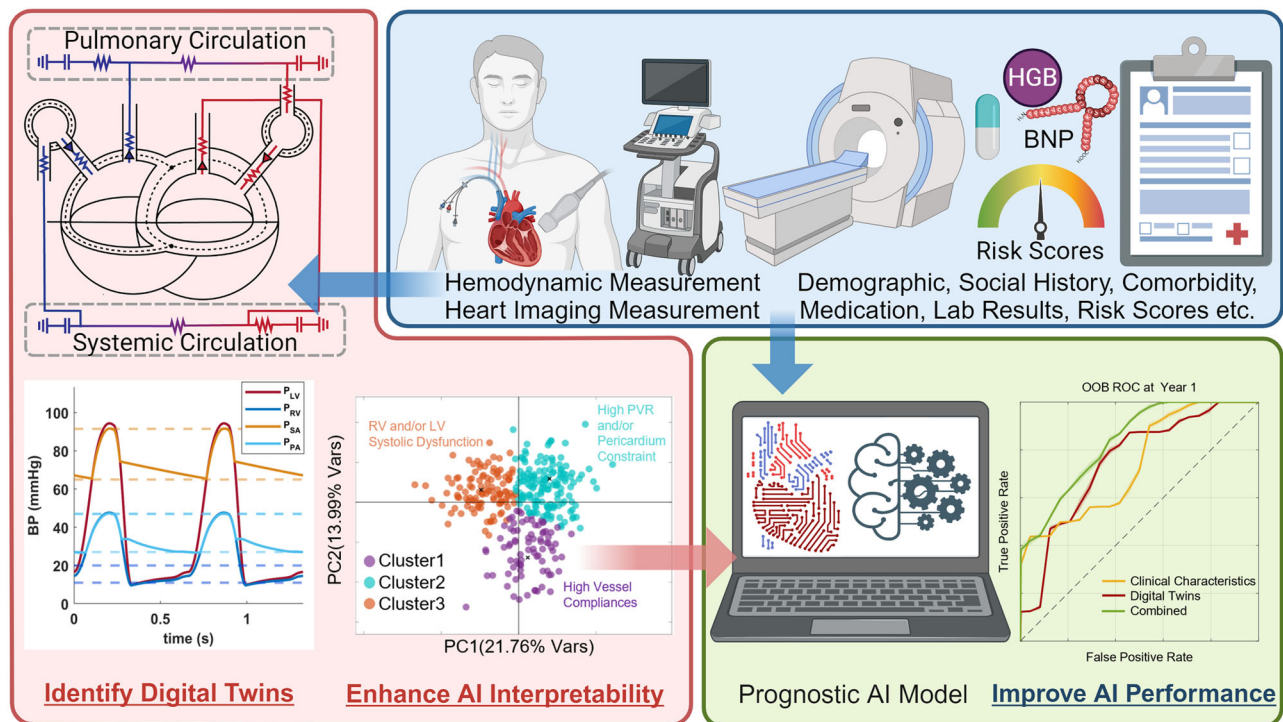


Fig. 8 | Overview of study workflow. The figure provides a summary of the schematic workflow and key findings from this study. Mechanistic computational models of the cardiovascular system were used to create digital twins, enabling the estimation of individualized parameters and simulations. These digital twins informed unsupervised machine learning to identify interpretable phenogroups and

mechanistic drivers of cardiovascular death risk. The integration of digital twin-derived features into prognostic AI models enhanced performance, transferability, and interpretability. This figure was generated using BioRender.com with permission under license agreement (Created in BioRender. Gu, F. & Schenk, N. (2025) <https://BioRender.com/138e240>).

which explains why many of them are classified in Phenogroup 2 or close to it. The prognosis data for the UW cohort are also consistent with those of the UMHS cohort: only 2 patients from the UW cohort died (as indicated by black circles), with one of these patients in Phenogroup 3, which is expected to have the worst outcomes. Nearly 50% of the UW cohort is in Phenogroup 1, consistent with the UW cohort's relatively favorable outcomes.

Since the phenotypes of patients represented in digital twins encapsulate physiological/pathological interrelationships among clinical data, the digital twin is a robust vehicle for incorporating domain knowledge into supervised ML. Here, we demonstrate how integration of digital twins in AI-based analyses of patient data improves performance, interpretability, and transferability of predictive models. Specifically, we find that integrating digital twin features with empirical measurements yields the identification of novel phenogroups and risk factors and improves survival prediction in HF. Critically, phenogroups, risk factors, and survival models identified this way are interpretable by the nature of the digital twin descriptors. Finally, predictive models trained with digital twin features are more transferable than models trained only with clinical data, as evaluated based on an independent prospective cohort.

By employing both unsupervised and supervised ML, we have successfully identified novel risk and protective factors, as well as predictors for outcomes in patients with HF. Factors derived from clinical characteristics, such as high BNP levels being associated with poor outcomes, align with prior knowledge and are not particularly novel. However, the factors identified from digital twins, which are often difficult or impossible to measure directly, represent promising new insights. First, both supervised and unsupervised ML revealed that high vascular compliance is associated with better outcomes, a potential prognostic factor that has not been well investigated³³. Second, passive stiffness of the ventricle emerged as a risk factor from unsupervised ML. While this factor cannot be measured directly, it can be assessed through indicators of diastolic dysfunction and is increasingly recognized for its prognostic value in both HFrEF and HFpEF³⁴.

Additionally, pericardial constraint was identified as a risk factor by both supervised and unsupervised ML. While researchers have recognized the potential role of the pericardium in the etiology of HFpEF²⁸, its significance in prognosis has been underexplored and warrants further investigation. Furthermore, some known but challenging-to-measure risk factors, such as LVEDP, were also identifiable through digital twins. This capability highlights the substantial application potential of our digital twin workflow. Simulation outputs not directly targeted during optimization can still provide estimates of real physiological variables. Digital twins, identified using a reduced set of targets, retain the full range of simulation outputs. While these outputs may exhibit reduced accuracy due to increased degrees of freedom, they represent a compelling direction for future exploration.

A significant limitation in demonstrating the added value of digital twins is the imbalance between the training and testing sets. The UMHS cohort includes 107 deceased patients out of 343 (31%), whereas the UW cohort has only 2 deceased patients out of 86 (2%). Because of this imbalance, we expanded the primary endpoint for the UW cohort to include patients with rehospitalizations, increasing the event occurrence rate to 39 out of 86 (45%). Despite this adjustment, there remains considerable heterogeneity between the cohorts. Specifically, the UW cohort patients, who also have exercise RHC results, are generally younger and healthier than those in the UMHS cohort.

Despite differences in cohort composition and outcome definitions, ML models trained on digital twin data outperform those trained on raw clinical data alone. This transferability is crucial for evaluating treatment efficacy across heterogeneous heart failure populations. As shown in Supplementary Tables 5–8, the imbalance in the dataset did not significantly impact the model's predictive ability, as evidenced by competitive F1-scores, despite a slight drop in performance between specificity and sensitivity. This method should be validated on a larger cohort in future work, although similar class imbalances are expected due to the intrinsic characteristics of the studied population and the purpose of the proposed algorithm. The

rationale for using two distinct heart failure datasets is that our models are designed to create digital twins representing heart failure of varying etiologies, replicating hemodynamic abnormalities (e.g., elevated intracardiac filling pressures) in specific heart failure patients. The transferability of the model to HFpEF patients is particularly significant given the pressing need to evaluate the efficacy of different therapies in development. In the short term, training the model to assess treatment benefit versus harm in real patients will facilitate more informed decision-making for future patients with similar digital twins. This rapidly evolving field is poised to enhance clinical trials in the future through the use of digital twins¹⁹.

Although we developed a risk prediction model (Extended *MAGGIC* score) in this study, our aim is not to claim superiority over the original *MAGGIC* score or suggest immediate clinical adoption. The *MAGGIC* score remains a widely validated tool that is easy to implement across large cohorts. Instead, our objective is to demonstrate the potential added value of integrating digital twins into AI applications.

Despite the promising findings, the current study has several limitations, as discussed above. To advance the digital twin approach, several enhancements should be prioritized. Future studies should focus on multicenter prospective cohorts to test the generalizability and applicability of this method to a broader HF population. Additionally, refinements to the model itself are essential. One aspect involves accommodating input data that lack CMR measurements, as these are often difficult to obtain in clinical practice. Another opportunity for enhancement is incorporating patient-specific differences in relaxation kinetics into the computational model, which may be particularly important in diastolic dysfunction^{35,36}. Delayed and incomplete LV relaxation are clinically relevant in HF and have been shown through computational simulation studies to subtly influence cardiac and valvular hemodynamics^{37,38}.

Furthermore, integrating longitudinal data into the digital twins would enable the simulation of more dynamic clinical scenarios, offering better insights into patient-specific disease progression and management. In addition to longitudinal data, incorporating exercise hemodynamic measurements to create exercise-state digital twins could prove particularly valuable. This approach would aid in the early diagnosis and management of HFpEF by providing a deeper understanding of how patients' cardiovascular systems respond to physical stress. These advancements have the potential to address the current study's limitations, paving the way for more robust and comprehensive applications of digital twins and AI in heart failure management.

Methods

Study design and patient data extraction

A retrospective cohort of 343 patients, selected from all patients with HF who visited the UMHS between June 1, 2009, and November 30, 2023, was identified using EHR data. Cohort discovery is summarized in Fig. 1. In brief, all adult HF patients without a single ventricle defect and with at least one recorded RHC, TTE, and CMR were pre-screened ($N=2727$). Qualified examinations were defined based on criteria from patients' EHR: RHC must report CO, HR, PADP, PASP, and PCWP; TTE must report IVSd, LVPWd, LVEF, LVIDd, LVIDs, LAd, and E/A ratio; CMR must report LVEDV, LVESV, RVEDV, and RVESV. A 3-month window was defined within which cardiovascular function was assumed to remain stable. From the pre-screened population, 371 patients with qualified RHC, TTE, and CMR examinations within the specified 3-month window were included in the final cohort (Supplementary Fig. 19). Further exclusions were made for patients who were not diagnosed with HF before the time window ($N=11$); patients who underwent a heart transplant or LVAD implantation within the specified 3-month window ($N=3$); patients with a history of heart transplant prior to the 3-month window with no evidence of new-onset HF ($N=7$); and patients with a significant shunt, indicated by an abnormal Qp/Qs ratio (defined as greater than 1.2 or less than 0.8), were identified in their RHC/CMR reports ($N=11$). Finally, clinical variables, including demographics, pre-existing medical conditions, medication history, hemodynamics measurements, heart imaging measurements, laboratory tests, and

outcome data (endpoints defined as LVAD implantation, heart transplant, or all-cause mortality), were collected from EHR and double-blind verified by our clinical team. Survival data beyond 5 years are right-censored at year 5 (Supplementary Fig. 20).

An independent validation cohort of 113 patients was obtained from the University of Wisconsin-Madison (UW) ('external testing set'). These participants presented to the UW-Madison Pulmonary Hypertension Clinic with worsening dyspnea (New York Heart Association Class II-III) and underwent clinically indicated invasive cardiopulmonary exercise testing. Each participant received a comprehensive resting transthoracic echocardiographic evaluation, a six-minute walk test, and a clinical assessment by a PH specialist. Of these, 86 patients had sufficient data (RHC, TTE, and CMR measurements) to identify digital twins. This cohort includes patients with different World Health Organization Pulmonary Hypertension (PH) groups: I ($n=9$), II ($n=49$), III ($n=6$), and IV ($n=5$), and 17 patients who presented with non-cardiac dyspnea (no rest or exercise PH). Following the 2022 ERS/ESC guidelines, 32 participants were diagnosed with HFpEF, defined as having a resting PCWP ≥ 15 mmHg and/or an exercise PCWP ≥ 25 mmHg (for patients with EF $\geq 50\%$, HF symptoms, and structural, functional, or biomarker evidence of cardiac abnormalities). The follow-up period was one year, with the composite endpoint defined as all-cause mortality and rehospitalization.

The UMHS study was approved by the University of Michigan Institutional Review Board (IRB protocol number HUM00243399). As the research involved only minimal risk to subjects, relied on identifiable private information for medical chart review, and did not adversely affect the rights and welfare of the subjects, a waiver of informed consent was granted. The UW study was approved by the University of Wisconsin-Madison institutional review board (IRB protocol number 2019-0535) and adhered to the Declaration of Helsinki guidelines. The requirement for informed consent was also waived due to the retrospective nature of the study and the de-identification of data, in accordance with institutional policies.

Identification of digital twins

The computational model used for simulating digital twins includes both a central and a peripheral part (Fig. 2a). The central part, representing the heart, is based on the TriSeg heart model of Lumens et al.³⁹, which provides an idealized geometrical framework for modeling ventricular dynamics. The peripheral part consists of lumped elements representing the peripheral circulation, atria, and pericardium, which are coupled to the central TriSeg model to make a closed-loop representation of the cardiovascular system. This model represents an optimal balance between maintaining relatively low computational complexity and achieving sufficient detail to reflect key anatomical features assessed by TTE and CMR. The peripheral part of the model, representing systemic and pulmonary circulation, is coupled to the TriSeg heart model using an approach based on the work of Kim et al.⁴⁰, forming a closed-loop model and enabling the simulation of blood flow and pressure dynamics throughout the cardiovascular system. For each patient, the model's functional parameters are optimized by minimizing the weighted squared error between the simulated outputs and the corresponding measured pressures in the ventricles and systemic and pulmonary circulations, as well as the measured volumes and flows in the cardiovascular chambers and vessels. The mathematical expression of the cost function is as follows:

$$\text{Cost} = \sum_{i=1}^n w_i \left(\frac{(\text{simulation}_i - \text{measurement}_i)^2}{\text{scale}_i} \right) \quad (1)$$

where simulation_i represents the computational model simulation result, measurement_i represents the corresponding clinical measurement, scale_i is the scaling factor (population mean of this measurement) for the value, and w_i is the weight for the value. To achieve this, we employ a combination of Genetic Algorithm (*ga*), *patternsearch*, and *fminsearch* optimization functions from MATLAB (MathWorks Inc.), chosen for their effectiveness in

navigating the nonlinear, multi-dimensional optimization landscape. The entire framework is developed in MATLAB R2024a (MathWorks Inc.). A detailed description of the computational model implementation is provided in Supplementary Note 1.

Unsupervised clustering

Functional parameters and simulated valve regurgitation fractions were used as input features ($n = 29$) for clustering. Features were standardized using MATLAB's built-in `zscore` function prior to clustering. Outliers (128 out of 9947) that exceeded ± 3 z-scores were capped at the maximum or minimum values within the non-outlier range. To determine the optimal number of phenogroups in the cohort, we employed K-means, hierarchical clustering, and Gaussian mixture model (GMM) clustering, optimizing using both the Davies–Bouldin index (DBI) and the gap statistic^{41,42}. The DBI measures the ratio of within-cluster dispersion to between-cluster separation, while the gap statistic compares the variance within clusters to that of a random distribution of mock data. In practice, we first examined the dendrogram from hierarchical clustering (Supplementary Fig. 21b). Given the high heterogeneity in HF, we then applied both the DBI and the gap statistic to optimize cluster numbers for $n = 1$ to $n = 25$ using K-means, hierarchical clustering, and GMM. The results suggested an optimal number of 3 or 4 clusters (Supplementary Fig. 21a). Upon further comparison using K-means and hierarchical clustering for $n = 3$ and $n = 4$, we found that 252 patients were consistently assigned to the same clusters for $n = 3$, while the number of inconsistent assignments increased for $n = 4$. Therefore, we selected $n = 3$ as the optimal number of clusters due to its balance between clustering performance and interpretability. After identifying the optimal number of clusters, we applied both K-means and hierarchical clustering, and selected the clustering solution that was consistent across both methods. For patients consistently clustered by both methods, we used K-means to identify cluster centers in a 2D principal component space. The remaining patients ($N = 91$), who were not consistently clustered, were then projected into the same principal component space. Using the built-in MATLAB function 'knnsearch,' they were assigned to the nearest cluster centers, yielding the final clustering result. The detailed methodology of clustering is provided in Supplementary Note 2. The 3D volcano plot⁴³ methodology was adapted to visualize differences across the three groups. Details of the design of 3D volcano plot are provided in Supplementary Note 3.

Prognostics AI model and validation

Features derived from clinical characteristics and digital twins were used to train two commonly employed survival prediction models: the Cox proportional hazards model, a linear statistical approach, and the RSF, a non-linear ML method. Features with more than 20% missing values were excluded. Subsequently, the remaining missing values were imputed using the k-nearest neighbors algorithm with $N = 5$. For the Cox proportional hazards model, we first conducted univariate analysis to pre-select features, followed by Lasso-Cox regression with tenfold cross-validation⁴⁴. The model was trained using data from the UMHS cohort and externally validated with the UW cohort. For the RSF model, we performed feature selection using variable hunting⁴⁵, followed by forward stepwise selection to minimize the loss function, defined as 1 minus the OOB C-index. Hyperparameters were optimized based on the same loss function. To assess performance and quantify uncertainty, the forest was trained 25 times. Similar to the Cox model, the RSF was trained on the UMHS cohort and validated using the UW cohort. Both models were developed in R 4.4.1 using the packages `randomForestSRC`, and `glmnet`^{44,46}. The predictive performance of the models was evaluated using the C-index, time-dependent ROC curves, and Brier scores. A detailed description of the prognostic model implementation is provided in Supplementary Note 4.

Statistical analysis

All statistical analyses were performed using GraphPad Prism 10.2.2. Continuous variables were analyzed via unpaired t tests and one-way ANOVA, followed by Tukey's multiple comparison test for post hoc

analysis. Categorical variables were compared using the chi-square test, and Bonferroni correction was applied for multiple comparisons. Correlations were assessed using Pearson's correlation coefficient. Statistical significance was defined as $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, and $****p < 0.0001$. A multivariable Cox proportional hazards model was also conducted in GraphPad Prism 10.2.2. Optimal cluster numbers were determined using the DBI and gap statistic, computed with MATLAB R2024a (MathWorks Inc.) built-in `evalclusters` function over a range of 1–25 clusters. The C-index, ROC curves, time-dependent ROC curves, and iAUC were calculated in R 4.4.1 using the `survAUC`, `survival`, and `survivalROC` packages.

Data availability

The source data supporting the findings of this study have been de-identified to remove any protected health information (PHI). The data for digital twin identification (Fig. 2) are available on GitHub: <https://github.com/beards-lab/TriSeg-Digital-Twins.git>. The data for machine learning analysis (Figs. 4, 6, and 7) are available on GitHub: <https://github.com/beards-lab/MachineLearningOnHFDData.git>.

Code availability

The code used in this study is openly accessible. For digital twin identification (Fig. 2), the code is available on GitHub: <https://github.com/beards-lab/TriSeg-Digital-Twins.git>. For machine learning analysis (Figs. 4, 6, 7), the code is available on GitHub: <https://github.com/beards-lab/MachineLearningOnHFDData.git>.

Received: 26 November 2024; Accepted: 6 February 2025;

Published online: 18 February 2025

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Acknowledgements

This work was supported by the National Institutes of Health under grants R01HL173346-01 and KL2TR002374-07, and the Michigan Medicine-PKUHS Joint Institute for Translational and Clinical Research. We also acknowledge support from the Michigan Biology of Cardiovascular Aging Postdoctoral Fellowship, which provided funding for the training and career development of Feng Gu during this study. Figure 8 was generated using BioRender with permission under a valid license: Created in BioRender. Gu, F. (2025) <https://BioRender.com/l38e240>.

Author contributions

F.G., S.L.H., and D.A.B. conceptualized and designed the study. F.G., A.J.M., N.A.S., S.Z., V.E.S. and F.J. collaborated to develop a methodology for identifying digital twins, building on B.E.C.'s work under the guidance of D.A.B. F.G., F.J., S.Z., T.C., and A.M. contributed to the design and review of the study cohort, supervised by S.H. and D.A.B. N.A.S. and V.E.S. provided critical input to the development of the digital twins model. D.U., R.L., X.H., and Y.T. offered feedback on the detailed methodology. F.G., A.J.M., and E.W. worked on the machine learning aspects with guidance from B.E.C., K.N., and D.A.B. F.R. conducted the external study and provided the validation dataset. F.G., A.J.M., and D.A.B. drafted the initial manuscript, which was finalized with contributions from S.L.H. (clinical aspects), K.N. (machine learning), F.R. (clinical aspects), and X.H. (statistics). All authors reviewed and approved the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41746-025-01501-9>.

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